```
import pandas as pd;
```

```
# create the csv with no header and 5 coloumns.load it with your own coloumn names and make the first column the index.
data = """101,Namit,19,CSE,85
102,Parth,20,AIML,90
103,Naman,19,CSE,78
104,Rahul,21,IT,88"""

with open("student.csv", "w") as f:
    f.write(data)
```

```
df = pd.read_csv(
    "student.csv",
    header=None,
    names=["ID", "Name", "Age", "Department", "Marks"],
    index_col=0
)

print(df)
```

```
      Name  Age Department  Marks
ID
101  Namit   19        CSE     85
102  Parth   20       AIML     90
103  Naman   19        CSE     78
104  Rahul   21         IT     88
```

```
# write a file using ; as the delimiter and -999 for missing values.read it so -999 become nan.
data = """Name;Age;Marks
Namit;19;85
Parth;20;-999
Naman;-999;78"""

with open("student.csv", "w") as f:
    f.write(data)
df = pd.read_csv(
    "student.csv",
    sep=";",
    na_values=-999
)
print(df)
```

```
    Name   Age  Marks
0  Namit  19.0   85.0
```

```
1  Parth  20.0    NaN
2  Naman   NaN   78.0
```

```
# Read the big csv into chunks of 2000 rows and compute the mean of one coloumn per key across all chunks.
total = {}
for chunk in pd.read_csv("Titanic-Dataset.csv", chunksize=2000):
    for key, group in chunk.groupby("Pclass")["Fare"]:
        total.setdefault(key, []).extend(group.tolist())
for key, fares in total.items():
    print(key, ":", sum(fares) / len(fares))
```

```
1 : 84.1546875
2 : 20.662183152173913
3 : 13.675550101832993
```

```
#export a dataframe with orient ="records ",read it back with pd.read_json,and check equality with .equlas.
df = pd.DataFrame({"Name": ["Parth", "Rahul"], "Marks": [90, 85]})
df.to_json("data.json", orient="records")
df2 = pd.read_json("data.json", orient="records")
print(df.equals(df2))
```

```
True
```

```
#write a short html table by handand load it with read.html.
html = """<table border="1">
<tr><th>Bank Name</th><th>City</th><th>ST</th><th>Closing Date</th></tr>
<tr><td>First Bank A</td><td>Springfield</td><td>IL</td><td>2020-03-13</td></tr>
<tr><td>Second Bank B</td><td>Dayton</td><td>OH</td><td>2019-11-01</td></tr>
<tr><td>Third Bank C</td><td>Austin</td><td>TX</td><td>2019-02-22</td></tr>
<tr><td>Fourth Bank D</td><td>Austin</td><td>TX</td><td>2018-07-19</td></tr>
</table>"""
with open("bank.html", "w") as f:
    f.write(html)

df = pd.read_html("bank.html")[0]
print(df)
```

```
       Bank Name         City  ST Closing Date
0   First Bank A  Springfield  IL   2020-03-13
1  Second Bank B       Dayton  OH   2019-11-01
2   Third Bank C       Austin  TX   2019-02-22
3  Fourth Bank D       Austin  TX   2018-07-19
```

```
t = '{"shop": "Fresh Mart", "open": true, "owner": null, "items": [{"name": "Rice", "price": 60, "qty": 10}, {"name": "Oil", "price": 150, "qty": 4}, {"name": "Tea", "
```

```
# load t into a python object and print its type.
import json
```

```
data = json.loads(t)
print(type(data))
```

```
<class 'dict'>
```

```
#Print the name of the second item and the number of items.
print(data["items"][1]["name"])
print(len(data["items"]))
```

```
Oil
3
```

```
#print the value of owner .what python is this .
print(data["owner"])
print(type(data["owner"]))
```

```
None
<class 'NoneType'>
```

```
# 4. Convert the Python object back to JSON text with indentation 2
result = json.dumps(data, indent=2)
print(result)
```

```
{
  "shop": "Fresh Mart",
  "open": true,
  "owner": null,
  "items": [
    {
      "name": "Rice",
      "price": 60,
      "qty": 10
    },
    {
      "name": "Oil",
      "price": 150,
      "qty": 4
    },
    {
      "name": "Tea",
      "price": 120,
      "qty": 6
    }
  ]
}
```